

Tobii Eye-tracking Manual: Boston College Human Neuroscience Lab

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Code of conduct

The Human Neuroscience Laboratory houses two Tobii Pro Spectrum systems.

Eye-tracking rooms

The two Tobii eye-trackers are located in McGuinn 405F and 405G. Access to these facilities is restricted and must be arranged through Jason Geller, Ph.D.

System/Hardware Overview

- This system is a screen-based eye tracker capturing gaze data at speeds up to 600 Hz. It provides superior data quality and is designed for extensive research into behavior and eye movements—from fixation-based studies to micro-saccades. It combines video-based pupil- and corneal reflection eye tracking with dark and bright pupil illumination modes.
- Two cameras capture stereo images of both eyes for robust, accurate measurement of eye gaze and eye position in 3D space, as well as pupil diameter. Sampling frequencies range between 60-600 Hz, with precision and accuracy between 0.01-0.06° (RMS), and 0.3°, respectively. The system provides binocular eye tracking with total system latency of less than 3 frames and a blink recovery time of 1 frame. The dual-camera system enhances data precision and robustness for head movement.
- The hardware is combined with Tobii Pro Lab software, which provides a visual user interface and supports researchers through all phases of an eye-tracking experiment—from test design and recording to analysis. Tobii Pro Lab facilitates the combination of eye tracking with biometric data streams, such as EEG, GSR, and ECG. For co-registration with other data sources, the software allows for a TTL marker at every stimulus onset to ensure precise data synchronization.

Components of the Tobii System

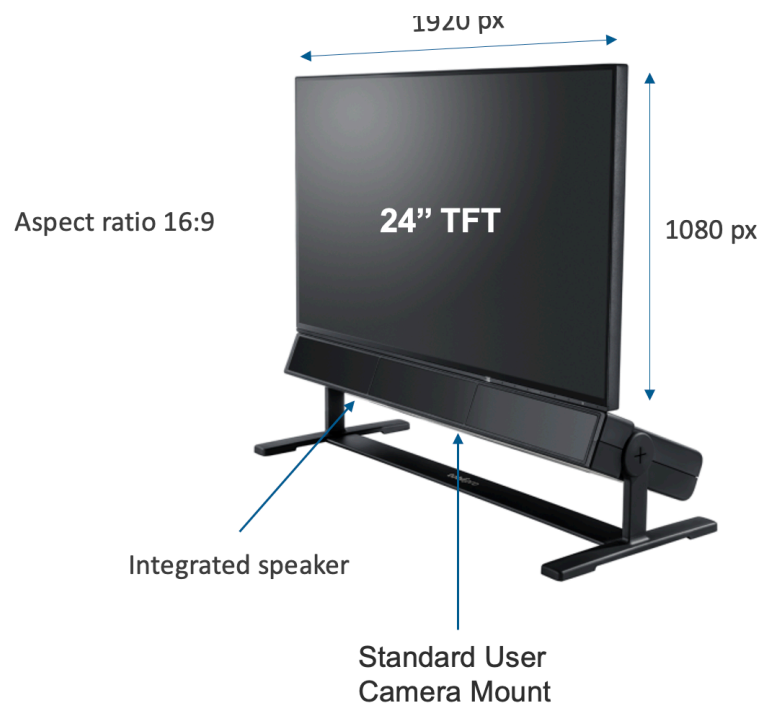
There are 4 critical components of the Tobii Spectrum systems:

1. An infrared light source
2. A high-speed video camera
3. The host monitor
4. The display monitor

- The Host PC and Display Monitor/eye-tracker are directly connected via an Ethernet cable.
- There are several advantages to separating eye tracking from stimulus control:
 - The Display PC can use all its resources to deliver stimuli with precision.
 - The Host PC can run a real-time OS and focus on tracking the eyes.
 - The Display PC can send commands and messages to the Host PC.

Computers

- Both 405F and 405G have identical setups:
 - Display monitor and eye-tracker:



i Note

The monitor is a 23.8-inch EIZO FlexScan EV2451 with a 16:9 aspect ratio, 1920 x 1080 pixel resolution, and a 5 ms gray-to-gray response time.

- Host PC:

- * The Host PC runs a real-time operating system for accurate timing to process the images, compute gaze location, parse the data, and store the data to disk.

i Note

Precision 3680 Mini Towers



- **Login Credentials:**

- **Username:** .\HNL_Lab_405
- **Password:** HN_Lab-mcguinn405

Additional Notes:

- **Data Management:** Please transfer eye-tracking data off the system as soon as possible. There is a research drive on the computer that data can be stored as a backup. Staff are not responsible for your data—this is solely the responsibility of the researcher.
- **Contact:** Contact Jason Geller when something is missing, broken, or when supplies are running low.

Who can use it?

- Contact the director for the rules, regulations, and access to the booking calendar and facilities.
- The Tobii trackers are available to BC faculty, students, and external users.
- Usage is not free [Costs].
- To use the system, you must complete training with the director.

Support/Training

- **Note:** Support generally does not include programming your experiment for you.
 - Templates for software like PsychoPy are available to help you program your experiment.
 - Troubleshooting assistance is provided if the experiment is not working properly.
 - **Preprocessing/Analyzing Data:**
 - * Tutorials are available to help with data preprocessing and analysis.
 - * Consulting is an additional cost.
 - * Helping you design your study, analyze data, or performing analyses/write-up results contributes to your project under the credit taxonomy <https://credit.niso.org/> and warrants authorship (regardless of payment).

Booking

- Book the lab via iLab. Ensure that the lab booking is under the appropriate schedule.
 - [Tobii Pro Spectrum system 1, 405G](#) (ilab link)
 - [Tobii Pro Spectrum system 1, 405F](#) (ilab link)

Costs

- You pay for each hour of testing:
 - **Self Use:** \$3.96/hr (Internal, External Non-Academic, External Academic)
 - **Assisted Use:** \$102.92/hr (needs approval)
 - **Training Session:** \$102.92/hr (needs approval)

File Storage

- On the desktops of both PCs, you will find an “Experiments” folder. Create a subfolder with your name and store all files there. Always copy your data to an external device and remove it from the PC.

! Important

The PC hard disks have limited space and will be cleaned regularly.

User Guide

Before the Participant Arrives

- Log in to iLab to start your session. Use the new kiosk system to turn on the equipment.
 - Navigate to **Go to Kiosk** under scheduling, find your reservation, and start your session.

Human Neuroscience Laboratory

About Our Core **Schedule Equipment** Staff Only Staff Only R

Schedule Resources

Go to Kiosk Timeline View

To make a reservation:

1. Click on the category name to expand the categories
2. Click on the instrument name or on the 'View Schedule' button on the far right.
3. On the calendar interface click and drag to reserve a time slot for the day and time desired
4. In the 'use and cost of reservation' section, use the drop down under *Pricing Details* button to select reservation type.
5. Click on the 'Save' button

Please be advised the calendars are for approved users only.

If you are requesting staff assistance or instrument training, after you create your reservation you will be notified via email if your reservation is approved/disapproved marked as 'non spam' to ensure they go to your inbox. Thanks!

64-channel BioSemi ActiveTwo system from Cortech Solutions, 405H [description](#) [pricing](#)

The laboratory houses a 64-channel BioSemi ActiveTwo system from Cortech Solutions. This system includes a compact ActiveTwo AD-box, into which sensor-signals attach. In addition to the 64 scalp-signals, the AD-box is channels from the face as well as from basic psychophysiological monitors. Each AD-box channel consists of a low noise DC coupled post-amplifier, with a first order anti-aliasing filter, followed by a Delta-Sigma modulator with a decimation filter with a steep fifth order sinc response and high resolution 24-bit output. The digital outputs of all the AD converters (up to 256) are digitally multiplexed and sent to the PC via a single optical fiber without a of data reduction.

There is a USB2 Receiver which converts the optical data coming from the AD-box to an USB2 output. This receiver also has a trigger port with 16 independent trigger inputs and outputs. This setup keeps the stimulation setu the subject. The trigger inputs allow easy setup of ERP measurements and event logging. The BioSemi headcaps (we have three different sizes of headcap to accommodate all adults) include the the electrode holders. This a active electrodes (approximately 15 minutes for a 64 channel headcap).

IF YOU ARE SCHEDULING USAGE FOR HUMAN SUBJECT RESEARCH PURPOSES, PLEASE ENSURE ALL PROCEDURES ARE IRB-APPROVED

Refresh Sessions

All Kiosk Sessions

Show completed sessions

Time Scale: -6 hours to 6 hours

Start typing to filter sessions

Clear

Owner	Resource	Start	End	Actions
Stephanie M Nemeh	Tobii Pro Spectrum system 1, 405G (Tobii Pro Spectrum system 1)	03 Oct 12:00 PM	03 Oct 01:30 PM	Start
Carolina Alvarez	Tobii Pro Spectrum system 1, 405G (Tobii Pro Spectrum system 1)	03 Oct 01:30 PM	03 Oct 04:30 PM	Start
Sandry Garcia	The Magstim Rapid ² repetitive transcranial magnetic stimulator, 405A (The Magstim Rapid ² repetitive transcranial magnetic stimulator)	03 Oct 03:00 PM	03 Oct 05:30 PM	Details

- Switch on both computers.

– Display/Eye-tracker:

- * There is a button on the back of the eye-tracker that must be pushed to turn it on (green means it is powered on).

- * If the monitor is off, use the switch on the back or the front monitor button to turn it on.

i Note

Make sure the Tobii is powered on for 10-15 minutes before recording eye data.

- **Host PC:**

- **Login Credentials:**

- * **Username:** .\HNL_Lab_405
 - * **Password:** HN_Lab-mcguinn405

3. Ensure the eye-tracker is connected:

- Open the Eye-tracking Manager on the desktop.
- Verify that the tracker is showing along with the serial number, address, and firmware version. If not, check the ethernet cable connections and try restarting the eye-tracker.

Upon Participant Arrival

i Difficulties with eyeglasses, drooping eyelids, long eyelashes, etc.

Eyeglasses

Some eyeglasses with anti-glare coating may reduce the amount of infrared light that reaches the eye.

Drooping eyelids or long eyelashes

Drooping eyelids or long eyelashes may occlude the pupil and affect tracking.

Contact lenses

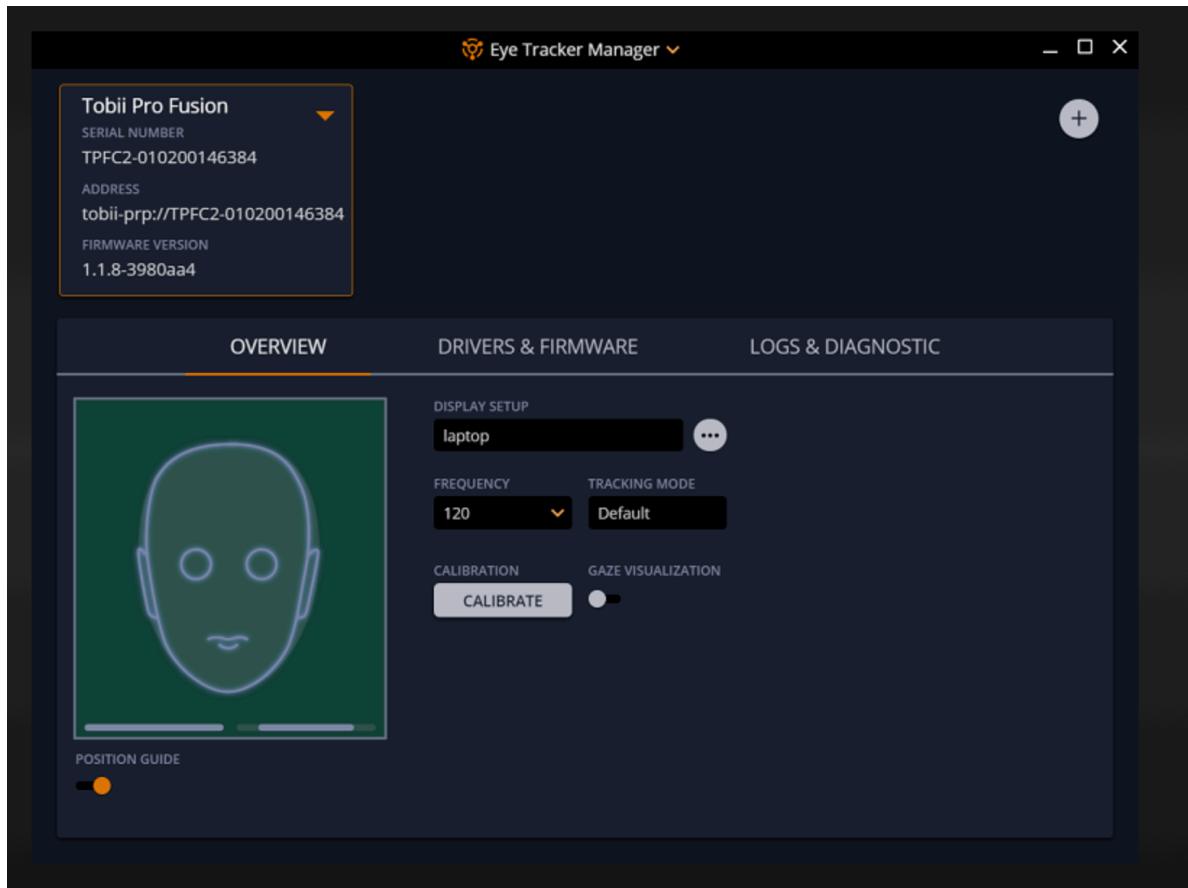
Soft contacts are generally fine, but hard contact lenses may not work well.

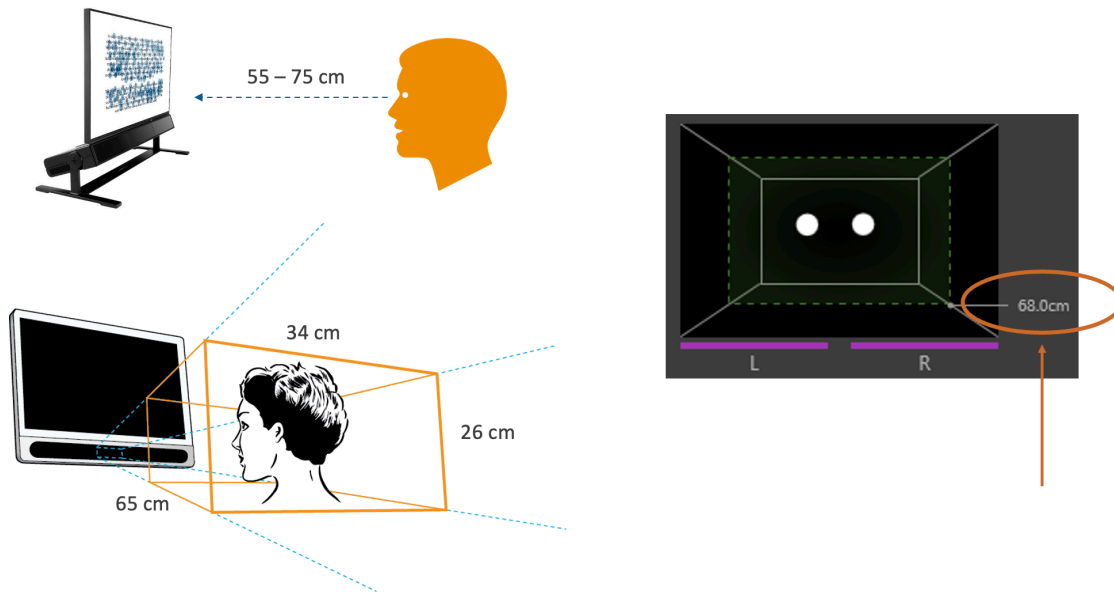
Makeup

Mascara or makeup that overlaps the pupil can disrupt tracking. You may ask participants to remove makeup during the study.

- Obtain consent.

- Ensure the participant is seated between 50–75 cm (20–37 inches) from the screen
 - Eye-tracker manager and Tobii Pro Lab have tools to help ensure adequate distance from the eye-tracker. Typically participants head should be located within the mock head (oval) if using eye-tracker manager and within the bounding box using Tobii Pro Lab. Some other software like (Titta and Psychopy) use a smiley face.





- Perform calibration and validation

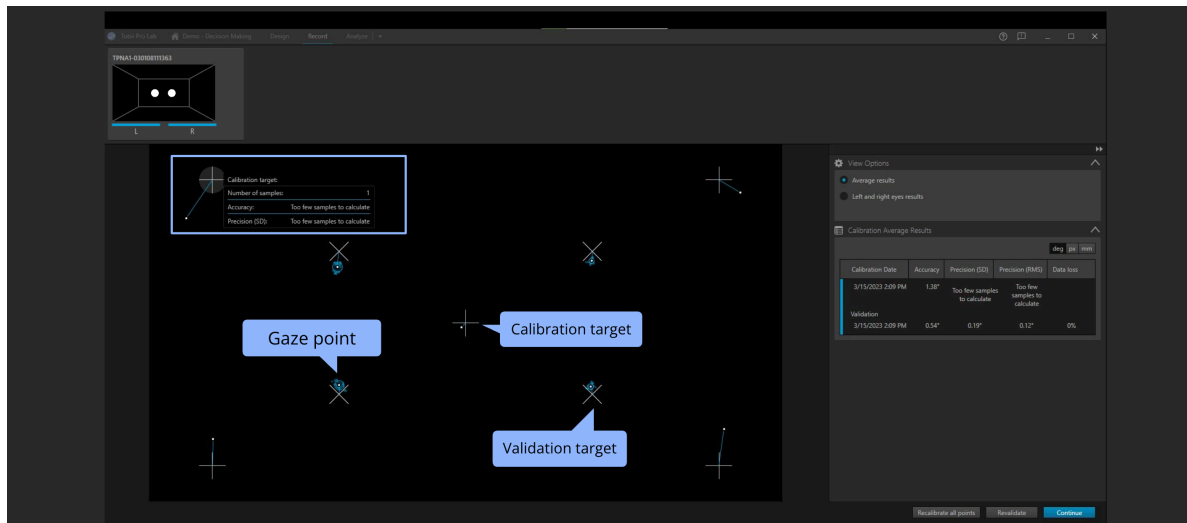
Calibration

Calibration is crucial for determining spatial accuracy and precision. The system displays several targets for the participant to fixate on. The calibration data will be used to compute gaze positions during recordings. This is usually done via Tobii Pro Lab or the experimental software of your choice.

Validation

Running validation after calibration helps you identify and correct any issues. The accuracy, precision (SD), and RMS are displayed after validation.

What qualifies as adequate depends on your study's goals. For instance, if you are studying reading, you might require both high accuracy (e.g., < 0.5 degrees) and high precision (e.g., $SD < 0.2$ degrees).



After Participant leaves

- Grant participant credit or pay them
- Turn off the display monitor and the eye-tracker
- Log off iLab to ensure correct time is logged for your study

Learning Resources

Tobii has a learning platform called Tobii Pro Academy (https://connect.tobii.com/s/welcome-to-tobii-pro-academy?language=en_US). I strongly encourage users to watch the videos before undertaking an eye-tracking experiment. You will need an account to watch the videos.

Software

There are a number of software that can be used in conjunction with the Tobii trackers

- PsychoPy
- Openseasme
- Tobii Pro Lab
- MatLab

The lab can provide most support with PsychoPy, Tobii Pro Lab and OpenSeasme. All of these softwares are on the lab computers.

Data Analysis Packages

- Pupil
 - R
 - * gazeR
 - * pupillometryR (<https://github.com/samhforbes/PupillometryR>)
 - * pupillometry (<https://github.com/dr-JT/pupillometry>)
 - Python
 - * pypillometry (<https://github.com/ihrke/pypillometry>)
 - Matlab
 - * Chap (<https://link.springer.com/article/10.3758/s13428-018-01190-1>)
- Reading
 - R
 - * popEye (<https://github.com/sascha2schroeder/popEye>) (does not work with Tobii)
- Fixation/gaze ROI
 - gazeR
 - webgazeR (webcam)

Further Readings

Eye Tracking Carter, Benjamin T., and Steven G. Luke. “Best Practices in Eye Tracking Research.” *International Journal of Psychophysiology: Official Journal of the International Organization of Psychophysiology*, vol. 155, Sept. 2020, pp. 49–62.

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Pupillometry Mathôt, Sebastiaan, and Ana Vilotijević. “Methods in Cognitive Pupillometry: Design, Preprocessing, and Statistical Analysis.” *Behavior Research Methods*, vol. 55, no. 6, Sept. 2023, pp. 3055–77.

Microsaccades Engbert, Ralf, and Reinhold Kliegl. “Microsaccades Uncover the Orientation of Covert Attention.” *Vision Research*, vol. 43, no. 9, Elsevier BV, Apr. 2003, pp. 1035–1045.

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